SAI ADITH SENTHIL KUMAR

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RESEARCH

AI Safety & Alignment · Mechanistic Interpretability · Model Evaluation · ML Systems

When Built-in Thinking Helps and Hurts | Sole-Author arXiv Preprint **2026**

- Sole-authored and executed a controlled study across 9 LLMs and 541 instruction-following tasks, using same-weight Thinking ON/OFF interventions to isolate the behavioral effects of built-in reasoning.
- Identified a Planning–Precision tradeoff where reasoning changed outcomes on up to 20.7% of prompts despite only 0.55–3.52 percentage-point aggregate accuracy differences, exposing systematic failure shifts hidden by aggregate metrics.
- Combined trace-relevance analysis with activation patching across 1.7B–14B parameter models, restoring up to 58% of reasoning-induced failures and identifying distinct internal behavior across constraint classes.

Counterfactual Instruction Following Benchmark | First-Author arXiv **Social Sim @ COLM 2025**

- Co-developed a benchmark for counterfactual instruction following, evaluating open- and closed-weight LLMs on reversed-performance personas and finding models remained up to 99% accurate despite explicit low-performance instructions.
- Found that one-shot prompting and self-refinement increased persona performance gaps by 72%; presented the work at COLM 2025 and received the 2025 GMU KEEN Research Award ($1,000).

Sparse Autoencoder Analysis of Persona-Specific Representations in LLMs | GitHub **2025**

- Designed a sparse autoencoder framework for mechanistic interpretability of counterfactual instruction following, analyzing Llama-3.1-8B and 70B models to identify persona-specific internal feature differences.
- Built a four-stage pipeline for activation extraction, L1-regularized SAE training, feature analysis, and cross-model comparison, scaling activation collection to 3,000 persona prompts with 4× data expansion to study counterfactual resistance.

Low-Overhead Model Instrumentation in vLLM | GitHub **2025**

- Developed runtime logit interventions and activation logging in vLLM for model-behavior and interpretability experiments, targeting ≤1% inference overhead.
- Designed a CUDA Graph-safe, three-stage asynchronous activation pipeline that sustained 97% of baseline inference throughput.

ENGINEERING EXPERIENCE

AI Engineer Intern | *NexGentic, Remote, Virginia* **Feb 2026 - Aug 2026**

- Improved multi-agent output accuracy from approximately 60% to 88% and reduced hallucinations from approximately 40% to 12% by developing LangGraph workflows with retrieval, critique, validation, and financial-analysis agents.
- Productionized Gemini and Vertex AI applications on Google Cloud Run with RAG, CI/CD, structured evaluation, and Model Armor guardrails for reliable enterprise deployment.

AI Research and Engineering Intern | *AriesView, Remote, Massachusetts* **May 2025 - Aug 2025**

- Reduced manual review time by 80% across 1,000+ contracts and document-processing time by 70% by developing a lease-abstraction system combining dual-stage OCR, retrieval, and LLM-based extraction.
- Improved retrieval precision by 90% and reduced hallucinations by 40% through chunking, reranking, and semantic search; deployed the containerized pipeline on Azure for 100+ concurrent users.

AI Engineer Intern | *Voicebotics AI, Remote, Florida* **Jan 2025 - May 2025**

- Achieved 95%+ insight relevance, reduced data latency by 30%, and improved meeting-conversion support by 40% by building a real-time recommendation system with OpenAI, FastAPI, WebSockets, and MongoDB.

Machine Learning Engineer Intern | *Rise Consultancy EDU, Remote, Virginia* **May 2024 - Aug 2024**

- Increased student success rates by 30% by deploying a quantized Llama 3-70B model with FastAPI and AWS SageMaker for adaptive learning-path generation.
- Improved model precision and recall by 20% through supervised fine-tuning, automated bi-weekly retraining with Airflow DAGs, and stratified k-fold cross-validation.

EDUCATION

George Mason University **Fairfax, VA**

M.S. Computer Science, Machine Learning Concentration **Aug 2026 - May 2027** | GPA: 3.89/4.00

B.S. Computer Science **Aug 2022 - May 2026** | GPA: 3.56/4.00

TECHNICAL SKILLS

Research: Mechanistic Interpretability, Sparse Autoencoders, Activation Patching, LLM Evaluation, Benchmark Design, Causal Interventions, Fine-Tuning, LoRA

ML / Systems: PyTorch, Hugging Face, vLLM, TRL, LangGraph, CUDA Graphs, Ray, MLflow, Docker, Kubernetes, Airflow

Languages / Cloud: Python, C++, SQL, Java, TypeScript; GCP, AWS, Azure